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NOZZLE OPTIMIZATION FOR SPACE-BASED VEHICLES

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ABSTRACT

The method of designing the exhaust nozzle contour for optimum thrust by variational methods is restricted to nozzle lengths above a certain minimum. This is referred to as the region of valid computation of the control surface where a shock-free solution is attainable. Below this minimum length, however, a solution that is shock-free is not possible by these methods. This is referred to as the invalid region. A solution for the boundary between the valid and invalid regions is obtained for the case of equilibrium and frozen chemistry. Also, a method for designing short axisymmetric optimum thrust nozzles in the invalid region that includes the effect of equilibrium and frozen chemistry is discussed. A discontinuity is introduced at a point on the control surface that crosses the nozzle exit flow. The discontinuity is achieved by a series of isentropic compression waves that coalesce at the point of discontinuity. Upstream of the control surface the flow is isentropic and shock-free. The boundary between the valid and invalid regions is defined for a typical equilibrium chemistry case and results are presented. Several nozzle contours are designed using this approach and the corresponding vacuum performance is presented.

NOMENCLATURE

a	= speed of sound
C_F	= vacuum thrust coefficient
$\%L$	= nozzle percent length, $\frac{X_E \tan 15^\circ}{R_T (\sqrt{\varepsilon} - 1)} 100$
M	= Mach number
M^*	= dimensionless velocity, V/a^*
R	= radial coordinate
V	= velocity
X	= axial coordinate
α	= Mach angle, $\sin^{-1} 1/M$
γ	= ratio of specific heats
ε	= nozzle area ratio
η_{DIV}	= divergence efficiency
θ	= flow angle

Subscripts

B	= tangency point
C	= axis point
D	= control surface point
E	= exit point
T	= throat point

Superscripts

*	= critical condition ($M = 1$)
'	= point in the invalid region

INTRODUCTION

Reducing nozzle length to save engine weight and minimize envelope dimensions while maintaining high levels of performance is a primary objective of nozzle design. This is especially critical for upper-stage engines and space-based propulsion systems where high area ratio nozzles are required.

It is well known that maximum thrust (ignoring viscous effects) is obtained from the nozzle by designing the contour according to the method of Rao¹. This method suggests a rapid expansion of the flow downstream of the throat section followed by a contoured nozzle that gradually turns the flow toward the axial direction. The nozzle contour from the throat to the exit is designed to produce isentropic flow with flow parameters compatible with the method of characteristics and mass continuity. Applying calculus of variations for maximum thrust to the flow parameters along the control surface, one computes the velocity, flow angle, and axial and radial location at every point along this line. The nozzle contour is then be computed by marching upstream from the control surface to the nozzle throat ensuring the flow parameters necessary for maximum thrust.

More specifically, this approach uses a suitably chosen right Mach line from point B as shown in Fig. 1. Using θ_B to define the Mach line and M^*_D to locate point D on the line, the optimum thrust control surface DE and the nozzle contour BE are defined. Moving point D along the chosen right Mach line toward the throat region results in lower M^*_D , higher θ_D , and a corresponding nozzle of lower area ratio and shorter length. However, below a limiting value

of M^*_D , the solution of the thrust optimization equations cannot be satisfied. This is a limitation of the method of Ref. 1. The point D corresponding to this limiting value of M^*_D defines the boundary of valid computation of the control surface on the chosen right Mach line. By choosing several right Mach lines, one can define a boundary for the region where valid computations can be carried out (i.e. the boundary of the valid region). All locations of point D closer to the throat are denoted to be in the invalid region since computation of the control surface cannot be accomplished.

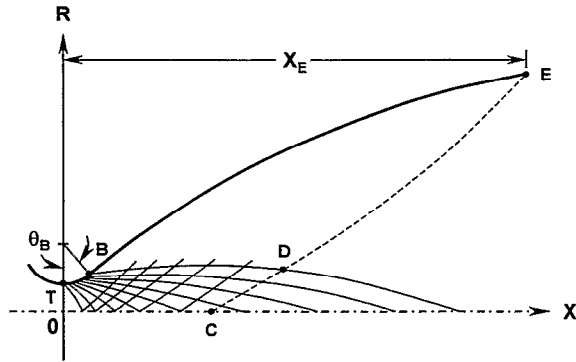


Fig. 1 Characteristic net and control surface.

Computation of the control surface depends on M^*_D as well as θ_D . By introducing a discontinuity in the flow parameters M^* and θ at point D', one can overcome the limitation of the method of Ref. 1 and still use variational methods to continue the control surface computations in the invalid region. This is accomplished by varying the amount of initial expansion and subsequent gradual compression, resulting in shorter length nozzles and accompanied by a small reduction in thrust. Such modifications have been considered in Ref. 2 for the case of a perfect gas (constant γ). The discontinuity in the flow parameters at point D' leads to discontinuous exit flow (DEF) nozzles.

This paper extends the method of Ref. 2 to include both equilibrium and frozen chemistry flows to better simulate the flow properties and extend the boundary of valid computations in more detail. The effect of introducing a discontinuity in the nozzle exit flow is examined in more detail. A nozzle design study is presented for a typical case of equilibrium chemistry to show the nozzle performance sensitivity to nozzle length and the impact of discontinuity in the exit flow.

OPTIMUM THRUST BOUNDARY

The boundary of the valid region along any right Mach line is defined in terms of the ability to compute the control surface. The boundary for the case of a perfect gas was examined in Ref. 2, resulting in the following relation between α and θ at point D.

$$\frac{\gamma - \cos 2\alpha}{\sin 2\alpha} \frac{\sin \theta}{\cos(\theta - \alpha) \cos \alpha} + \frac{\tan(\theta - \alpha)}{\tan \alpha} = 1$$

In cases where the flow cannot be treated as a perfect gas, it is possible to consider equilibrium or frozen chemistry and derive a relation similar to that above.

In optimization of thrust of a nozzle with a condition of fixed length, it was shown in Ref. 1 that the flow parameters along the control surface DE are required to satisfy the following relation.

$$M^* \frac{\cos(\theta - \alpha)}{\cos \alpha} = \text{constant} = M^*_D \frac{\cos(\theta_D - \alpha_D)}{\cos \alpha_D} \quad (1)$$

Taking logarithms of both sides of Eq. (1) and differentiating, the above relation leads to the following equation.

$$\frac{dM^*}{M^*} - (d\theta - d\alpha) \tan(\theta - \alpha) + d\alpha \tan \alpha = 0 \quad (2)$$

It was also shown in Ref. 1 that the control surface DE is a left Mach line and that the flow parameters along the line satisfy the compatibility equation.

$$\frac{dM^*}{M^*} = d\theta \tan \alpha + \frac{dR}{R} \sin \alpha \tan \alpha \frac{\sin \theta}{\sin(\theta + \alpha)} \quad (3)$$

The second term on the right-hand side of Eq. (3) controls the ability to compute the control surface DE. By solving Eqs. (2) and (3) along with the characteristic equation [Eq. (4)] starting from point D, the control surface is constructed to point E.

$$\frac{dR}{dX} = \tan(\theta + \alpha) \quad (4)$$

In the above equations, $d\theta$, $d\alpha$, dX , and dR are finite differences between successive points along the control surface. Calculations are carried out with a finite increment $\Delta\theta$ to obtain the values of M^* , α , R , and X at successive points along the control surface

DE until the integrated mass flow matches that through BD. However, when the simultaneous solution of Eqs. (2) and (3) does not yield a positive radial increment dR further computations are not possible (i.e. point D is in the invalid region).

A simplified procedure to determine the boundary for valid computation of the control surface is presented in the following discussion. Substituting Eq. (3) into Eq. (2) results in the following relation.

$$d\theta[\tan(\theta - \alpha) - \tan \alpha] - d\alpha[\tan(\theta - \alpha) + \tan \alpha] = \frac{dR}{R} \sin \alpha \tan \alpha \frac{\sin \theta}{\sin(\theta + \alpha)} \quad (5)$$

The left-hand side of the Eq. (5) is evaluated at every point along the chosen right Mach line from point B. When this function becomes zero, the right-hand side of Eq. (5) leads to dR equal zero to denote that point D is on the boundary. Rearranging the terms in Eq. (5) results in the following condition to be satisfied between θ and α at point D to continue the computations of the control surface and construction of the optimum thrust contour.

$$1 - \frac{d\alpha}{d\theta} \left(\frac{\tan(\theta - \alpha) + \tan \alpha}{\tan(\theta - \alpha) - \tan \alpha} \right) \geq 0 \quad (6)$$

VALID AND INVALID REGIONS FOR POINT D

To identify the boundary of valid computation of the control surface, a right Mach line and valid locations of point D are discussed for a typical case of equilibrium oxygen/hydrogen chemistry. Geometric parameters X and R and flow parameters M^* and θ at various locations along the right Mach line corresponding to $\theta_B = 42^\circ$ are shown in Figs. 2 and 3.

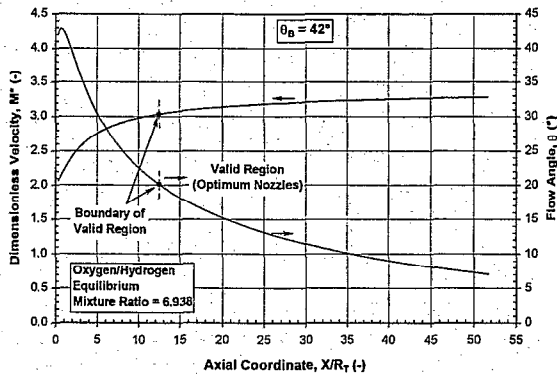


Fig. 2 Flow properties along right running Mach line corresponding to $\theta_B = 42^\circ$.

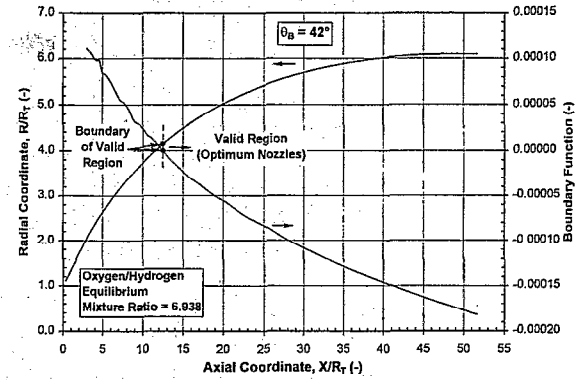


Fig. 3 Boundary function along right running Mach line corresponding to $\theta_B = 42^\circ$.

In using Eq. (6) for locating point D on the boundary of the valid region, it should be noted that θ and α are flow parameters and $d\theta$ and $d\alpha$ are finite differences between successive points along the control surface. In computing the control surface, a finite difference $\Delta\theta$ is specified along DE to obtain M^* and α from Eq. (1). It is necessary to note that $dM^*/d\theta$ and $d\alpha/d\theta$ along the control surface will have opposite signs to those occurring along the right Mach line shown in Figs. 2 and 3. To obtain positive values of dR along DE, it is necessary to have the left-hand side of Eq. (6) greater than zero at point D. The left-hand side of Eq. (6), denoted the boundary function, is also calculated and its variation with axial coordinate is shown in Fig. 3. When this function reaches zero, dR must be zero since the rest of the terms in the right-hand side of Eq. (5) are non-zero. Point D is on the boundary of the valid region for computing optimum thrust contours for the right Mach line corresponding to $\theta_B = 42^\circ$ as shown in Fig. 4. For all locations on and to the right of this boundary point, one can obtain optimum thrust contours according to the method of Ref. 1.

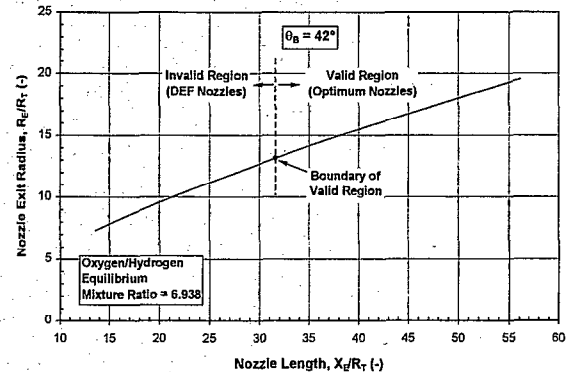


Fig. 4 Coordinates of point E in the valid and invalid regions corresponding to $\theta_B = 42^\circ$.

USE OF INVALID REGION

The point on the boundary of the valid region as given by Eq. (6) is shown in Fig. 4 for the right Mach line corresponding to $\theta_B = 42^\circ$. Further reduction in nozzle length can be achieved by using a lower value of $M^*_{D'}$ on the chosen right Mach line which moves point D into the invalid region denoted as point D'. Discontinuities in both M^* and θ at the point D' are introduced by considering a bundle of isentropic compression waves coalescing at point D'. The Prandtl-Meyer relation controls the amount of flow compression at the point such that Eq. (6) is satisfied. The isentropic compression coalesces into an induced shock wave at point D' but the flow process upstream of the control surface and point D' is still shock-free and isentropic. Nozzle optimization is performed using the modified flow parameters at point D' to compute the DEF nozzle contour. Several such nozzles have been computed using this method and the non-dimensional coordinates of the nozzle exit point are shown in Fig. 4 along with the optimum thrust nozzles in the valid region.

DISCONTINUOUS EXIT FLOW NOZZLE MAP

Similar to optimum thrust nozzles designed in the valid region, DEF nozzles designed in the invalid region can also be mapped. The nozzle design map presents nozzle area ratio as a function of nozzle percent length for families of $M^*_{D'}$ and θ_B . Such a nozzle design map is shown in Fig. 5 for a range of $M^*_{D'}$ and θ_B . This map shows a family of DEF nozzles with a range of area ratio from 40 to 250 and a range of nozzle percent length from 46 to 76. The boundary of the valid region is shown as the dashed line and indicates the minimum length limit of optimum nozzles designed in the valid region. The DEF nozzle design methodology extends the nozzle design space to shorter lengths than previously possible.

As shown in Fig. 5, nozzle area ratio increases with increasing length for a fixed value of $M^*_{D'}$ or a fixed value of θ_B . The method of Ref. 1 for designing optimum thrust nozzles is limited to designs on and to the right of the boundary of the valid region, whereas the DEF design methodology extends the nozzle design space to the left of the boundary.

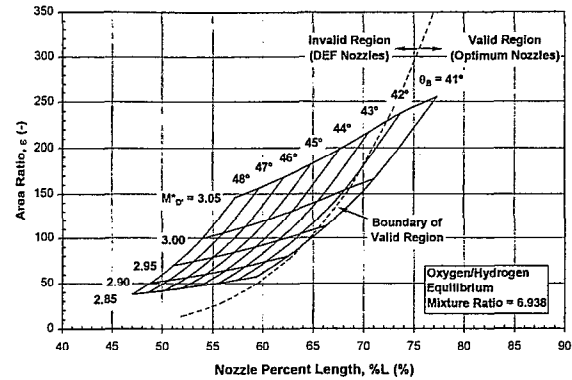


Fig. 5 Nozzle design map in the invalid region.

NOZZLE STUDY INCLUDING OPTIMUM AND DEF NOZZLES

Vacuum thrust coefficient, excluding viscous effects, as a function of nozzle percent length is shown in Fig. 6 for a family of DEF nozzles of different area ratio. These nozzles were designed using the design map of Fig. 5. For comparison, the thrust performance of a corresponding family of optimum thrust nozzles designed in the valid region is also included. As expected, a reduction in nozzle length is accompanied by a decrease in thrust. Also evident is the increase in thrust performance as nozzle area ratio increases from 50 to 250. Presenting the results in this form clearly shows the distinction between the valid and invalid region for optimum thrust nozzle design.

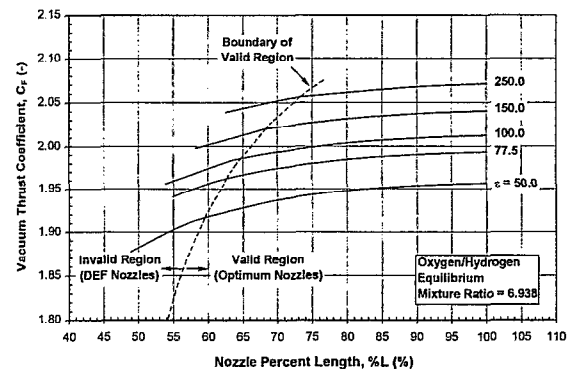


Fig. 6 Vacuum thrust coefficient as a function of nozzle percent length and area ratio.

Presenting the results in terms of a more physical feature of the nozzle, vacuum thrust coefficient is shown in Fig. 7 as a function of non-dimensional nozzle length X_E/R_T . These results clearly show an upper limit on vacuum performance as a function of nozzle length. This upper limit appears to coincide with the boundary of the valid region. The nozzle

that delivers maximum vacuum thrust for a given length is obtained by selecting the area ratio on the boundary of the valid region. It is also noted that a considerable drop in thrust performance occurs as nozzle length is reduced. For nozzles that are not strictly limited to vacuum operation, other design factors must be considered (e.g. performance and flow separation at sea level). Another observation made from the results shown in Fig. 7 is that the same level of vacuum performance can be achieved with a shorter nozzle by increasing area ratio.

The corresponding divergence efficiencies of the DEF and optimum thrust nozzles are shown in Fig. 8. As expected, the divergence efficiency diminishes as nozzle length decreases for a specified area ratio.

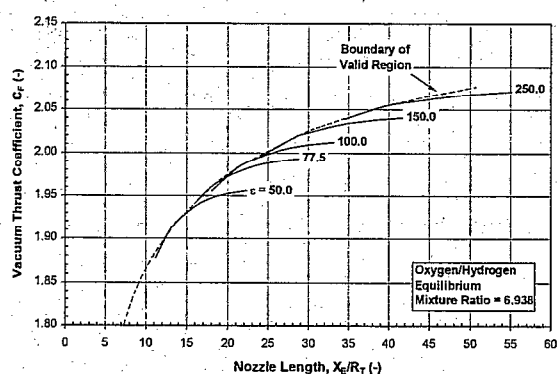


Fig. 7 Vacuum thrust coefficient as a function of nozzle length and area ratio.

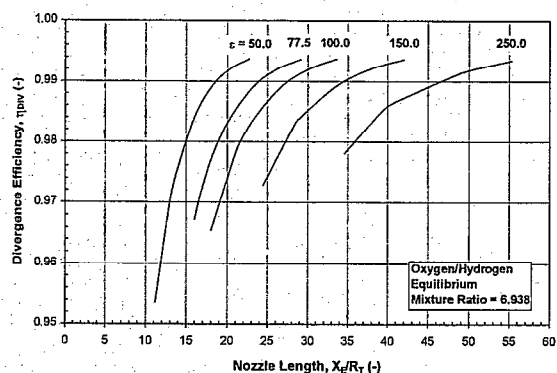


Fig. 8 Divergence efficiency as a function of nozzle length and area ratio.

CONCLUSIONS

Nozzles of shorter length were designed for a typical case of equilibrium oxygen/hydrogen chemistry. The design space for optimum thrust nozzles was extended to allow shorter nozzles to be considered in nozzle design studies. The boundary of the region of valid control surface computations and the invalid region were defined. A method for designing contours in the invalid region was developed for the case of equilibrium and frozen chemistry. An isentropic compression is introduced at point D' on the control surface to affect a change in the flow properties M^* and θ such that the thrust optimization equations are satisfied. The flow upstream of the control surface is isentropic and shock-free.

The results of a nozzle design study are presented showing thrust performance as a function of nozzle length and area ratio. The boundary between the valid and invalid regions on a typical nozzle design map was clearly established, and it was shown that the boundary coincides with the upper limit of vacuum performance for any given nozzle length. Divergence efficiency as a function of nozzle length was determined for a set of different area ratios. It was clearly shown that nozzle efficiency decreases significantly for shorter nozzle lengths. The results also showed that a specified vacuum performance can be achieved by a number of nozzle length and area ratio combinations, and that shorter lengths drive the nozzle to higher area ratios until the limiting condition is reached at the boundary of the valid region.

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